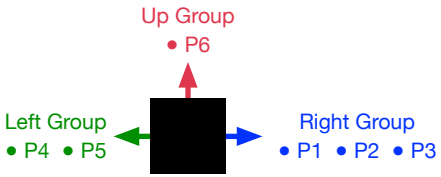
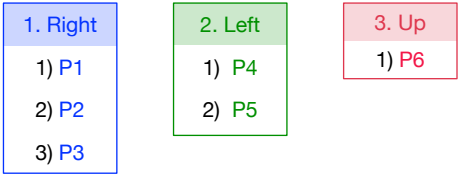
1. Group paths by primary direction

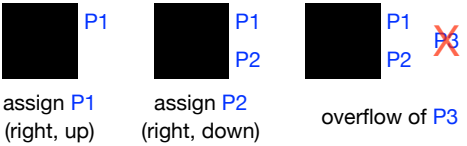
PathID	Preference Order (cost)			
P1	right: 0.1	up: 0.6	down: 1.3	left: 1.8
P2	right: 0.3	down: 0.7	up: 0.9	left: 1.6
P3	right: 0.4	up: 0.8	down: 1.1	left: 1.9
P4	left: 0.1	down 0.5	up: 1.2	right: 2.0
P5	left: 0.2	up: 0.3	down: 1.5	right: 1.8
P6	up: 0.1	right: 0.5	left: 1.2	down: 1.7



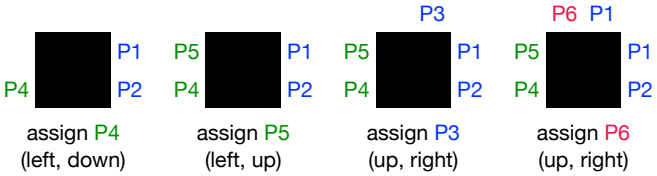
2. Order groups by size and paths by cost



3. Assign paths by secondary direction



4. Overflow Handling

P3 is assigned to up group, cost set to zero

5. Process is repeated until all paths have been assigned